Zirui Zhao | Curriculum Vitae

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Education

Computer Science **School of Computing, National University of Singapore**
Ph.D., Singapore *2020 - 2024*
Advisor: Prof. David Hsu

Automation(Qian Class) **EECS & Qian Xuesen College, Xi'an Jiaotong University**
B.Eng., Xi'an, China *2016 - 2020*
Advisor: Prof. Pengju Ren
Thesis: Robustify Route-based Visual Place Recognition by Bridging Kalman Filter with CNN Representation

Visiting

Safe AI Lab **Carnegie Mellon University**
Summer intern, Pittsburgh, USA *2019 Summer*
Advisor: Prof. Ding Zhao

School of Computing **National University of Singapore**
Visiting student, Singapore *2018 Summer*
Advisor: Prof. Soo Yuen Jien

Publications & Manuscripts

[2]: R. Chen, W. Wang, **Z. Zhao**, D. Zhao, "Active Learning for Risk-Sensitive Inverse Reinforcement Learning ," submitted to ICRA 2020, preprinted in arXiv.

[1]: **Z. Zhao**, Y. Mao, Y. Ding, P. Ren and N. Zheng, "Visual-Based Semantic SLAM with Landmarks for Large-Scale Outdoor Environment," 2019 2nd China Symposium on Cognitive Computing and Hybrid Intelligence (CCHI), Xi'an, China, 2019, pp. 149-154.

Selected Researches & Projects

Safe AI Lab **CMU**
Active Risk-sensitive Inverse Reinforcement Learning *2019 Summer*
Risk-sensitive inverse reinforcement learning provides an general model to capture how human assess the distribution of a stochastic outcome when the true distribution is unknown (ambiguous). This work enables an RS-IRL learner to actively query expert demonstrations for faster risk envelope approximation.

Cognitive Architecture Lab **XJTU**
Visual Semantic SLAM *Nov. 2018 - Jun. 2019*
This work combines the semantic labels with pointclouds generated from ORB-SLAM by using bayesian update rules. Besides, the landmarks in the real world is attached in the map, associated with the pointclouds by using fuzzy mathematics.

Cognitive Architecture Lab **XJTU**
Multi-robot Cooperative navigation *Apr. 2018 - Mar. 2019*
This project established a multi-robot navigation and exploration system, which consist of UAVs and UGVs.
Demo video link: https://www.youtube.com/watch?v=wSB4hyW9rWc&t=5s

Selected Scholarships & Honors

2020 - 2024: NUS Research Scholarship

2020: XJTU Outstanding Graduate Honor

2019/2018/2017: XJTU Excellent Student Honor

2018: XJTU 2nd Prize of 1989 Mechanical Engineering Alumni Scholarship for Qian Xue Sen Program

2017: XJTU Siyuan Merit Scholarship

Interests & Skills

Interests: Cognitive Robotics, Robot Planning under Uncertainty, Probabilistic Robotics, Human-Robot Interaction.

Language: C++, Python, MATLAB, LaTeX

Tools: TensorFlow, PyTorch, Keras, Scikit-Learn, Numpy, Jupyter, OpenCV, ROS.

Software: Carla, GitLab, GitHub.